A Universal Physical Consumer-Resource Neural ODE Framework for Microbial Serial Dilution Ecology

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1. Background: Serial Dilution Regimes & Modeling Dilemma

- **Serial Dilution Ecology:** Cycles of continuous intra-cycle nutrient uptake and metabolic interaction interrupted by periodic discrete dilution ($1 : D$) and dispersal (μ).
- **Why Generalized Lotka-Volterra (gLV) Fails:** Requires N^2 empirical interaction coefficients, assumes fixed interaction signs, cannot model substrate depletion, and overfits severely ($N = 48$ requires 2,304 parameters).
- **The Simplex Reset Artifact:** Normalizing relative abundances to $\sum x_i = 1$ at transfers mutes quadratic self-limitation ($-\alpha_i N_i^2$) and triggers artificial competitive exclusion cascades.

2. Physical Consumer-Resource Neural ODE Formulation

We model continuous within-passage growth coupled to discrete dilution transfers:

$$\frac{dN_i}{dt} = N_i(t)\left(\sum_{k=1}^{K} \Theta_{ik} R_k(t)\right) - \alpha_i N_i(t)^2$$

$$\frac{dR_k}{dt} = -R_k(t)\left[\sum_{i=1}^{N} \Theta_{ik} N_i(t) - \sum_{l=1}^{K} W_{lk}\left(\sum_{j=1}^{N} \Theta_{jl} N_j(t)\right)\right]$$

- **Physical Constraints:** $\hat{} \geq 0$ (uptake traits), $\mathbf{W} \geq 0$ (cross-feeding), $\alpha > 0$ (self-limitation), $\mathbf{R}_0 = \text{softplus}(\mathbf{a}_c) \geq 0$.
- **Physical Continuity Across Passages ($m \to m+1$):** $N_i(0)^{(m+1)} = \frac{N_i(\tau)^{(m)}}{D} + \mu \cdot M_{c,i}, \quad R_k(0)^{(m+1)} = R_{0,c,k}$. Eliminates simplex resets.
- **Closed-Form Effective Interaction Operator:** $\mathbf{A}^{(c)} = -\text{diag}(\alpha) - \kappa\, \hat{}\, \text{diag}(\mathbf{R}_{0,c})(\mathbf{I} - \mathbf{W})^{T}\hat{}^{T}$. Decomposes into intra-specific crowding ($-\alpha$), niche competition ($-\hat{}\mathbf{R}_0\hat{}^{T}$), and facilitation ($+\hat{}\mathbf{R}_0\mathbf{W}^{T}\hat{}^{T}$); rank $\leq K \ll N$.

3. Out-of-Sample Generalization: Venturelli et al. (2018)

- **Held-Out Test Consortia:** 73 complex synthetic human gut communities (4,380 dynamic points) never seen during model training.
- **Minimal Validation Loss Architecture ($K = 7$):** Optimal trade-off between expressive capacity and generalization ($\mathcal{L}_{\text{GKL}} = \mathbf{0.12950}$).

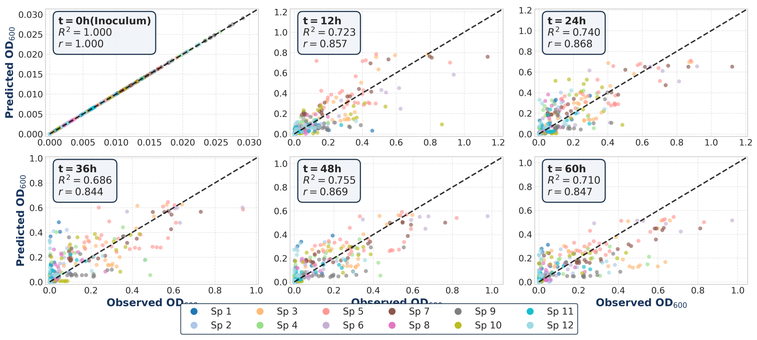


Figure 1. **Six-Panel Held-Out Generalization Across Time ($t = 12n$, $n = 0 \ldots 5$).** Observed vs. predicted absolute biomass (OD_{600}) across 73 unseen gut communities and 12 species from $t = 0$h to 60h ($R^2 = \mathbf{0.69}$–$\mathbf{0.75}$, Pearson $r = \mathbf{0.84}$–$\mathbf{0.87}$ across dynamic timepoints; overall test $R^2 = \mathbf{0.7233}$, $\log_{10} r = \mathbf{0.9652}$, Spearman $\rho = \mathbf{0.9901}$).

4. Universality of Metabolic Trait Matrices: Pollard et al. (2023)

- **Spatial Carrier Microenvironments:** 50 human gut isolates transferred across 5 carriers: Liquid broth, Mucin carriers, Plain agar, and Supernatants.
- **Universal Invariant Traits:** A single metabolic uptake matrix $\hat{}$ (50×5) and cross-feeding matrix $\mathbf{W}$ (5×5) govern all 5 environments. Carrier divergence is controlled strictly by the learned environmental vector $\mathbf{a}_c \in \mathbb{R}^5$.
- **Spatial Refugia Prevent Washout:** Solid carriers stably preserve **21–24 species** ($\exp(H) \approx 11$–15) over 100 serial passages by sheltering sensitive taxa, whereas liquid broth collapses to 11.0 species ($\exp(H) = 5.44$).

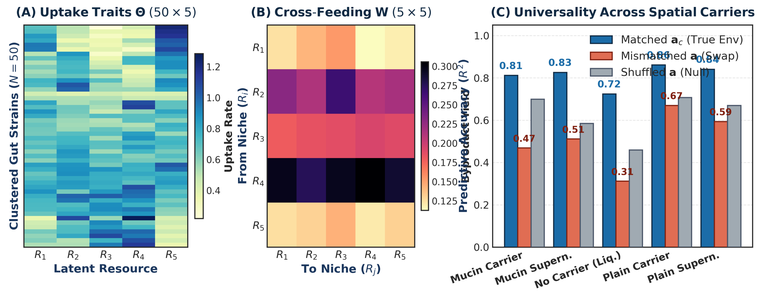


Figure 2. **Universality Across 5 Spatial Microenvironments. (A)** Clustered uptake traits $\hat{}$ reveal specialized metabolic guilds. **(B)** Cross-feeding byproduct matrix $\mathbf{W}$. **(C)** Matched $\mathbf{a}_c$ achieves $R^2 = \mathbf{0.725}$–$\mathbf{0.862}$ (mean $\mathbf{0.813}$), whereas carrier swaps ($R^2 = 0.511$) and shuffled nulls ($R^2 = 0.580$) collapse.

5. Meaningful Environmental Vectors: German et al. (Rumen)

- **Longitudinal Methanogen Recovery ($N = 91$ Strains, Peak $\geq 1\%$):** Retains **22 methanogenic archaea** capturing **98.5**% of archaeal reads and **92.3**% of total community coverage across 6 geochemical perturbations.
- **Environmental Specificity vs. Severe Null Collapse:** True learned $\mathbf{a}$ achieves $\mathbf{R^2 = 0.8224}$ (Pearson $\mathbf{r = 0.9074}$, $\mathcal{L}_{\text{KL}} = \mathbf{1.7128}$ at $K = 12$). Condition swaps cause a **59.7**% collapse in accuracy ($R^2 = 0.3313$), while shuffling $\mathbf{a}$ completely destroys predictive ability ($R^2 = -1.1257$).

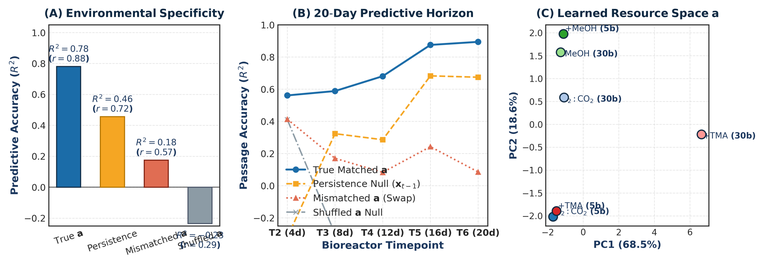


Figure 3. **Environmental Specificity vs. Null Baselines. (A)** True learned $\mathbf{a}$ ($R^2 = \mathbf{0.822}$, $r = \mathbf{0.907}$) significantly outperforms persistence null ($R^2 = 0.559$), mismatched conditions ($R^2 = 0.331$), and shuffled null ($R^2 = -1.13$). **(B)** Multi-passage horizon stability over 20 days. **(C)** PCA of learned $\mathbf{a}$ vectors autonomously clusters substrate and pressure chemistry.

6. Phase Space Dynamics & Multi-Cycle Resource Divergence

- **Non-Equilibrium Bifurcations (Gore et al., *Science* 2022):** 48 isolates across 99 communities. Minimalist $K = 2$ model captures transitions between stable fixed points and persistent chaos **purely by modulating $\mathbf{R}_0$**.
- **Identical Community, Divergent Phases ($S = 12$, Community 3):** Species pool $\mathbf{m}$ is 100% identical; switching $\mathbf{R}_{0,\text{LN}} \to \mathbf{R}_{0,\text{HN}}$ triggers chaos ($P = 0.0108 \to 0.8921$).

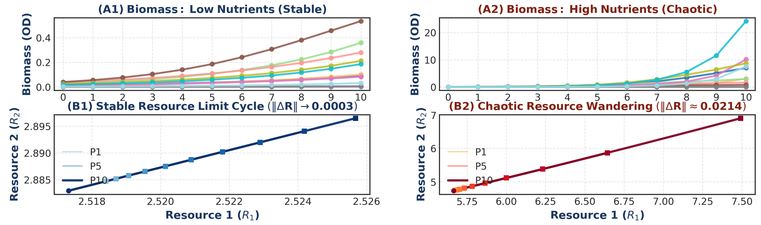


Figure 4. **Stability vs. Chaos Driven Purely by Environmental Resources $\mathbf{R}_0$. (A1–A2)** Biomass dynamics $N_i(t)$ damping to a fixed point in LN vs. irregular oscillations in HN. **(B1–B2)** Multi-cycle resource trajectories (R_1, R_2) converge onto an identical closed limit cycle in LN ($\|\Delta\mathbf{R}\| \to \mathbf{0.0003}$), but wander and diverge in HN ($\|\Delta\mathbf{R}\| \approx \mathbf{0.0214}$).

7. Exploring Model Extremes: Taxonomic Biomarkers of Chaos

- **Locked 100% Recall Chaos Classifier:** Trained on $K = 2$ physical trajectory features ($F_1 = 0.7458$, Recall = 100%, $T^* = 0.3891$). Correctly separates experimental $S = 24$ communities in Medium Nutrients into stable and chaotic phases.
- **Chaotic Drivers (Sp 31–36, +80% in Chaos):** Generalists with balanced traits ($\theta_1 \approx \theta_2$) that concentrate competitive pressure in the niche center, driving chaos.
- **Stabilizing Specialists (Sp 37–42, +80% in Stable):** Extreme niche partitioners ($\theta_1/\theta_2 > 1.5$) that cleanly divide resources and enforce stability (a taxonomic mechanism missed by bacteria-agnostic random matrix models).

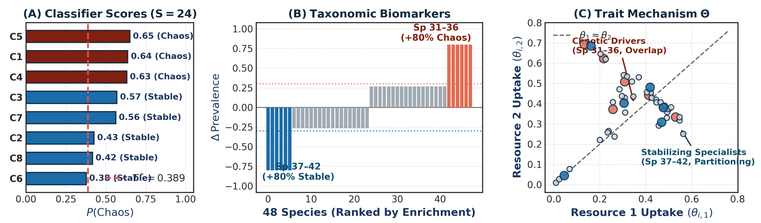


Figure 5. **Classifier Separation & Taxonomic Biomarkers of Chaos. (A)** Classifier scores on experimental $S = 24$ communities in MN. **(B)** Species enrichment ΔPrevalence. **(C)** Metabolic trait map in $\hat{}$ space.

8. Key Takeaways & Conclusions

1. **Trait Universality:** Metabolic traits ($\hat{}$) and cross-feeding networks ($\mathbf{W}$) are universal biological invariants across spatial carriers and chemical perturbations ($R^2 = 0.72$–0.97).
2. **Meaningful Environment Encoding:** Learned vectors $\mathbf{a}_c$ autonomously capture genuine environmental chemistry, outperforming null baselines and persistence models.
3. **Deterministic Chaos & Biomarkers:** Transitions between stability and chaos can be modulated purely by initial resources $\mathbf{R}_0$. Chaos is driven by symmetric niche generalists, whereas specialized resource partitioners enforce ecological stability.

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